

La Independiente: Designing AI Ubiquitous Systems for Latin American and Caribbean Women Crowdworkers

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Figure 1: Example of our intelligent system that guides women crowdworkers to build community and increase their skills.

ABSTRACT

Since 2018, Venezuelans have contributed to 75% of leading AI crowd work platforms' total workforce [15], and it is very likely other Latin American and Caribbean (LAC) countries will follow in the context of the post covid-19 economic recovery [7]. While crowd work presents new opportunities for employment in regions of the world where local economies have stagnated[11], few initiatives have investigated the impact of such work in the Global South through the lens of feminist theory. To address this knowledge gap, we surveyed 55 LAC women on the crowd work platform Toloka to understand their personal goals, professional values, and hardships faced in their work. Our results revealed that most participants shared a desire to hear the experiences of other women crowdworkers, mainly to help them navigate tasks, develop technical and

soft skills, and manage their finances more efficiently. Additionally, 75% of the women reported that they completed crowd work tasks on top of caring for their families, while over 50% confirmed they needed to negotiate their family responsibilities to pursue crowd work in the first place. These findings demonstrated a vital component lacking from the experiences of these women was a sense of connection with one another. Based on these observations, we propose a system designed to foster community between LAC women in crowd work to improve their personal and professional advancement.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing.**

KEYWORDS

crowd-work, gig work, AI labor, Latin American and Caribbean women, wearable technology

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INTRODUCTION. In recent years, Latin American and Caribbean (LAC) workers have been identified as significant crowd work contributors. The crisis conditions in these areas have left many dependent on crowd work as a steady source of income. LAC women make up a sizable portion of workers using crowd work as a path to financial independence while balancing traditional caregiver responsibilities expected of them due to their strong patriarchal societies. Unfortunately, research surrounding perspectives of LAC women is largely incomplete as most investigations center on the experiences of Western women, who comparably have greater freedom and support to prioritize their needs[2].

Designing a space for crowdworkers to share knowledge can improve their work quality by providing a community where they can seek support. According to field practitioners and researchers, a better future in crowd work depends on whether transparency, fair pay, and opportunities for professional advancement are embodied in tools available to them [10]. Having their experiences documented will benefit workers long-term and allow them to support one another in identifying areas of improvement in their work environments. This can inform decisions to organize and advocate for respect, fair pay, and transparency from requesters and crowd work platforms [12].

Our research is guided by feminist theory, meaning it is grounded in a set of theoretical traditions that privilege women's issues and experiences [9]. A fundamental particularity of feminist research lies in its focus on power imbalance in the subject studied and the relationships between the subject and the research. Feminist research practice includes utilizing reflexivity or positionality as a tool for producing knowledge and as an ethical method. It focuses on subverting unequal power relationships in knowledge production by considering our position within the research as one of power and always considering the vulnerability of the researched [8], which in this study are the LAC crowd workers.

METHODS. Our team conducted a survey in Spanish on the global crowd source platform Toloka to understand the characterization of LAC women crowdworkers. To gain insight into how gender affected their work, we developed a set of 60 questions based on widely recognized gendered factors such as time poverty [5], work-home balance, outside attitudes to the work done, care work, and direct opinions about whether they had considered gender as a liability to their work. Although Toloka only permits workers to identify on a gender binary scale, we included more inclusive options for participants to indicate their gender identity in our study. Our survey was divided into 6 sections: most questions were Likert scale and open-ended. We exclusively deployed the survey to workers in the LAC region using Toloka's geographic filters. All participants received compensation for completing the survey.

RESULTS. This article aimed to understand the characterization of LAC women crowdworkers regarding their motivations, needs, socioeconomic background, skills, and hopes. Our survey study included 60 participants (55 female, 3 male, 2 preferred not to say) from 12 LAC countries. The sample represented four age groups,

with 76% of respondents being 18 to 35 years old. The majority answered that their gender had little or no impact on their experience performing crowd work, while 17% believed it had some or a critical impact. Of the respondents in the age group, most had a professional degree; 18% held a bachelor's degree, while 13% had a master's degree. Participants shared they worked about 2 times per week from their own devices at home and spent about 20 minutes completing a common task. The job they performed most was data labeling, web page testing, and surveys (only 10% performed audio transcription tasks). Income per task was in the range of \$0.01 to \$0.16. 91% of the respondents conducted other economic activities in addition to Toloka. The most recurring reason they performed crowd work was to gain supplementary income. Additionally, a preference for working from home and having a flexible schedule were essential factors. Few people considered it a recreational activity, and one respondent mentioned crowd work as one of the few possibilities for employment. Regarding the geopolitical situation of their country of origin, participants considered the main impacts to be economic, which reduced their professional prospects and required them to perform a second economic activity; 21% remained neutral. Most importantly, the respondents shared their preferred features for a tool to aid them in their work. There was a high agreement that the explicability of tasks is critical. Although crowdworkers in Toloka can contact the requester with questions, multimedia resources and tutorials are helpful for them to understand the tasks fully. Similarly, 63% of respondents use search engines and translating applications for their work, so prioritizing language learning features in the tool design can significantly help them develop their skills and gain independence from external tools, thus working more efficiently. Regarding the perceived need to share experiences with other workers, the main goal of a communication feature would be to increase task completion, profitability, and performance.

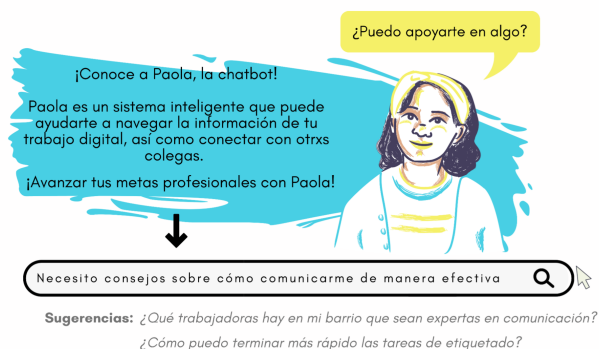


Figure 2: Example of the intelligent chatbot users would interact with to navigate the platform and assist them in finding information and other workers relevant to their questions and interests.

DISCUSSION. In recent years, the rise of gig economy platforms has led to increasing number of women participating as crowdworkers, offering their skills and services remotely. This scientific paper presents a comprehensive survey conducted with women crowdworkers from Latin America and the Caribbean to delve into their

labor practices, challenges, and gain valuable insights to inform the design of a supportive and inclusive system. Our research aims to shed light on the unique difficulties faced by women in the gig economy and devise effective solutions to improve their working conditions. For our research, we take inspiration from prior work which has studied the experiences of different types of crowdworkers [4], especially those in rural areas and within Latin America [1, 6, 13]. However, more work is needed to conduct studies with this population properly.

DESIGN IMPLICATIONS. The general dynamics of crowd work (relatively low wages, depersonalized work, and asymmetric power relationships [16]) have led researchers to raise ethical concerns that we are building a future on the backs of exploited workers in digital sweatshops. To address these shortcomings, a “Tooling Add-on” Movement has emerged that makes tools to transform gig platforms and convert them into spaces where we would be proud to have our children work [3, 10, 14]. Based on these ideas, we propose to design an AI-powered social connection and recommendation system specifically designed to assist crowdworkers in building a supportive network and developing their technical and soft skills. The system’s main feature would be a chatbot that recommends social connections to a user based on their expressed interests to facilitate long-term relationships between workers. We will model the system according to the design preferences derived from culturally influenced work values in LAC (e.g., emphasis on community and extensive options to navigate the platform based on occupational role). In addition, the chatbot will be available in the multiple languages spoken in LAC (Spanish, Haitian Creole, English, etc.), allowing them to navigate the feature in their native language or practice their skills as needed. We recognize it is critical not to treat LAC as a monolith. Each design decision should be carefully considered after detailed participatory design sessions, or the presentation should be adjustable based on user preferences. Fig 1. and Fig 2. present an example of our design implications. We also aim to integrate collaboration features directly into the recommender system. One example comprises a wearable device that monitors users’ behaviors and health states to recommend when they should take breaks. It encourages them to connect with other workers who were also notified to rest by their own devices. These social breaks can facilitate interactions they usually miss by the nature of working independently by emulating the “water cooler chats” shared by workers in office jobs. Another opportunity to foster collaboration exists when recommending tasks based on the user’s history and preferences. At this time, it will also show them an online post or group discussing the best ways to approach the assignment, thus promoting communities they can join to receive advice. Collaborative recommender tools enable seamless networking and skill building without disrupting users’ workflows.

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REFERENCES

- [1] Walter Ángel, Saiph Savage, and Nataly Moreno. 2015. Participatory stoves: Designing renewable energy technologies for the rural sector. In *Proceedings of the 18th ACM Conference Companion on Computer Supported Cooperative Work & Social Computing*. 259–262.
- [2] Janine Berg, Uma Rani, et al. 2021. Working conditions, geography and gender in global crowdwork. *Work and Labour Relations in Global Platform Capitalism*. Cheltenham: Edward Elgar (2021), 93–110.
- [3] Chun-Wei Chiang, Anna Kasunic, and Saiph Savage. 2018. Crowd coach: Peer coaching for crowd workers’ skill growth. *Proceedings of the ACM on Human-Computer Interaction* 2, CSCW (2018), 1–17.
- [4] Claudia Flores-Saviaga, Yuwen Li, Benjamin Hanrahan, Jeffrey Bigham, and Saiph Savage. 2020. The challenges of crowd workers in rural and urban America. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, Vol. 8. 159–162.
- [5] Laura M. Giurge, Ashley V. Whillans, and C.D. West. 2020. Why time poverty matters for individuals, organisations and nations. *Nature Human Behaviour* 4, 10 (Aug 2020), 993–1003. <https://doi.org/10.1038/s41562-020-0920-z>
- [6] Benjamin V Hanrahan, Ning F Ma, Eber Betanzos, and Saiph Savage. 2020. Reciprocal research: Providing value in design research from the outset in the rural united states. In *Proceedings of the 2020 International Conference on Information and Communication Technologies and Development*. 1–5.
- [7] Karen Hao. 2023. How the AI industry profits from catastrophe. *MIT Technology Review* (Jul 2023). <https://www.technologyreview.com/2022/04/20/1050392/ai-industry-appen-scale-data-labels/>
- [8] Donna Haraway. 1988. Situated Knowledges: the science question in feminism and the privilege of partial perspective. *Feminist Studies* 14, 3 (Jan 1988), 575. <https://doi.org/10.2307/3178066>
- [9] Sharlene Nagy Hesse-Biber. 2013. *Feminist Research practice: A Primer*. SAGE Publications.
- [10] Aniket Kittur, Jeffrey V. Nickerson, Michael Bernstein, Elizabeth Gerber, Aaron Shaw, John Zimmerman, Matt Lease, and John Horton. 2013. The Future of Crowd Work. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work* (San Antonio, Texas, USA) (CSCW ’13). Association for Computing Machinery, New York, NY, USA, 1301–1318. <https://doi.org/10.1145/2441776.2441923>
- [11] V Nickerson, Jeffrey. 2013. Crowd work and collective learning. <https://ssrn.com/abstract=2246203>
- [12] Niall O’Higgins and Luis Caro. 2022. Crowdwork for young people: Risks and opportunities. (2022).
- [13] Pedro Reynolds-Cuellar, Marisol Wong-Villacres, Karla Badillo-Urquiola, Mayra Donaji Barrera Machuca, Franceli L Cibrian, Marianela Ciolfi Felice, Carolina Fuentes, Laura Sanely Gaytán-Lugo, Vivian Genaro Motti, Monica Perusquia-Hernandez, et al. 2023. Para Cima y Pa’Abajo: Building Bridges Between HCI Research in Latin America and in the Global North. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [14] Saiph Savage, Chun Wei Chiang, Susumu Saito, Carlos Toxtli, and Jeffrey Bigham. 2020. Becoming the super turker: Increasing wages via a strategy from high earning workers. In *Proceedings of The Web Conference 2020*. 1241–1252.
- [15] Florian Alexander Schmidt. 2019. Crowdsourced production of AI Training Data: How human workers teach self-driving cars how to see. *ideas.repec.org* (2019). <https://ideas.repec.org/p/zbw/hbsfof/155.html>
- [16] Carlos Toxtli, Angela Fuller, and Saiph Savage. 2020. Reputation Agent: Prompting Fair Reviews in Gig Markets. In *Proceedings of The Web Conference 2020*. 1–5.